

Frequency-domain training data for a learned plasma fluid closure

Tighter errors, $100\times$ cheaper data, same NN

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Setup: cold-bulk + warm-beam two-fluid

3-moment beam fluid coupled to cold bulk via Poisson:

$$\begin{aligned}\partial_t u &= -E, \\ \partial_t E &= u - \varepsilon(u_b U_0 + U_1), \\ \partial_t U_n &= -iku_b U_n - ikU_{n+1} + nM_{n-1}E.\end{aligned}$$

Closure problem: need U_3 from (U_0, U_1, U_2) .

Kinetic linear response (Maxwellian beam):

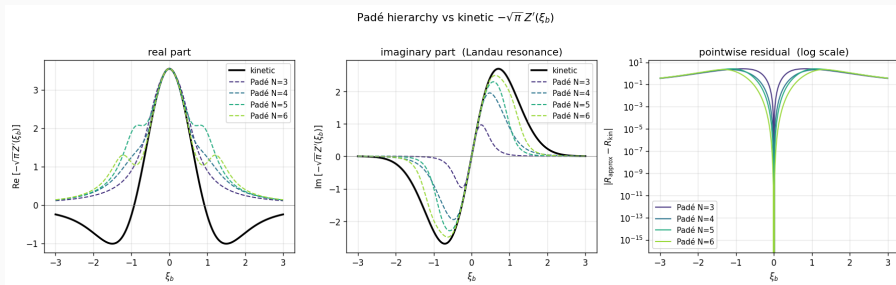
$$U_0/\hat{\Phi} = Z'(\xi_b), \quad \xi_b = \frac{\omega - ku_b}{kv_b}$$

Exact closure ratio:

$$\alpha(\xi_b) \equiv U_3/U_0 = \xi_b^3 - \xi_b/Z'(\xi_b)$$

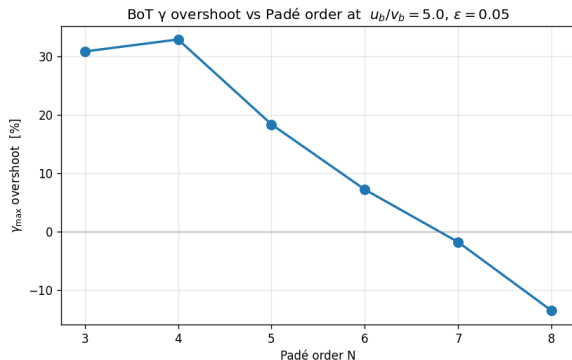
Padé baseline (HP / Hunana hierarchy)

Linear-in-moments closure $U_3 = \sum_i a_i U_i$ produces a degree-(1, 3) rational of $Z'(\xi_b)$. Hunana et al. 2019: extended hierarchy up to 8 poles.



Residual $\sim 10^{-2N+3}$ near origin, plateau at $\sim 10^0$ for $|\xi_b| > 1$: a fixed-order rational cannot match an entire function everywhere.

Padé overshoot vs hierarchy order



Canonical $(u_b/v_b, \varepsilon) = (5, 0.05)$:

N	overshoot
3	+30.8%
4	+32.9%
5	+18.4%
6	+7.2%
7	-1.8%
8	-13.6%

Crosses zero near $N=7$: classic non-monotone behaviour of rational approximation to a non-rational target.

A structural identity: the per-mode closure in closed form

Before introducing the NN, one closed-form identity worth recording — it determines what any closure must reduce to on a single eigenmode. The recurrence on a Maxwellian beam:

$$U_{n+1} = \xi_b U_n - n M_{n-1}, \quad M_0 = 1, \quad \boxed{M_1 = 0}.$$

Apply it three times:

$$U_1 = \xi_b U_0$$

$$U_2 = \xi_b U_1 - M_0 = \xi_b^2 U_0 - \quad (\text{since } M_0 = 1)$$

$$U_3 = \xi_b U_2 - 2M_1 = \xi_b U_2 \quad (\text{since } M_1 = 0)$$

Divide $U_3 = \xi_b U_2$ by U_0 and use $\xi_b = U_1/U_0$:

$$\boxed{\alpha = \frac{U_3}{U_0} = \frac{U_1}{U_0} \cdot \frac{U_2}{U_0}}$$

Exact per-eigenmode, zero parameters. Verify directly: $r_1 r_2 = \xi_b (\xi_b^2 - 1/Z'(\xi_b)) = \xi_b^3 - \xi_b/Z'(\xi_b) = \alpha(\xi_b)$. All the Z' machinery is absorbed into U_2 ; the closure step itself is complex multiplication.

Used directly in time-domain it fails: a superposition of N eigenmodes gives

$(\sum_i A_i U_1^{(i)}) (\sum_j A_j U_2^{(j)}) \neq (\sum_k A_k U_0^{(k)}) (\sum_\ell A_\ell U_3^{(\ell)})$ — cross-terms break it. So this identity is not a usable time-domain closure. *It's a constraint on what any per-mode closure must equal.* We will use it in two slides to ask: *is this what the NN learned?*

NN closure: replace the linear closure

Replace $U_3 = \sum_i a_i U_i$ (constants) by

$$U_3 = f_{\theta}(U_1/U_0, U_2/U_0) \cdot U_0$$

with f_{θ} a small MLP. Nonlinear-in-moments, no rational-order ceiling.

Architecture (fixed throughout). 4-layer MLP, (64, 64, 64, 64) hidden, tanh, $\sim 8.7\text{k}$ params. 4 reals in (re/im of two complex moment ratios), 2 reals out (re/im of α).

Question: where does the training data come from?

Default: train on simulation trajectories

Standard recipe in data-driven closures:

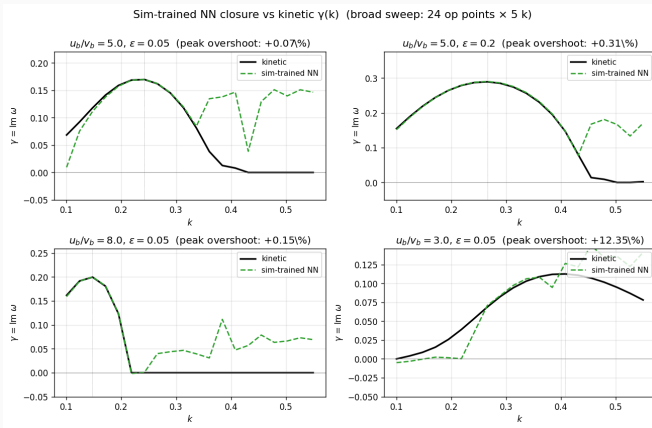
- Sweep $(u_b/v_b, \varepsilon, k, \text{IC})$, integrate kinetic linear system in time.
- At each time step record (U_0, U_1, U_2, U_3) along trajectory.
- Regress U_3/U_0 on $(U_1/U_0, U_2/U_0)$.

Broad training sweep.

- $u_b/v_b \in \{3, 4, 5, 6, 8, 10\}$
- $\varepsilon \in \{0.02, 0.05, 0.10, 0.20\}$
- 5 values of k per op pt, 12 random ICs each
- $\sim 10^5$ training samples after transient filtering

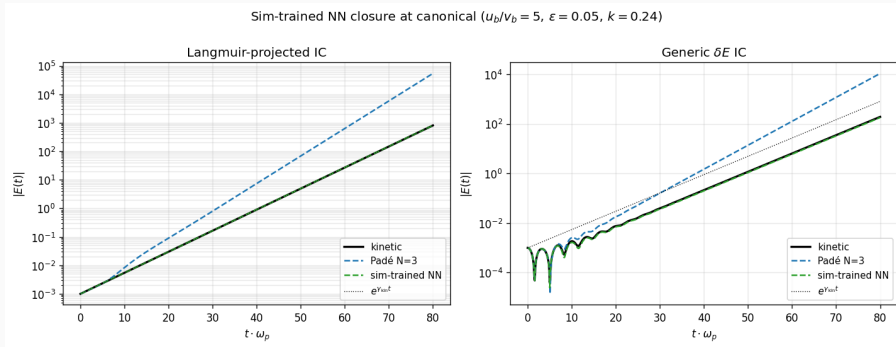
Designed to be *broad* for maximum coverage.

Sim-trained NN: dispersion looks fine



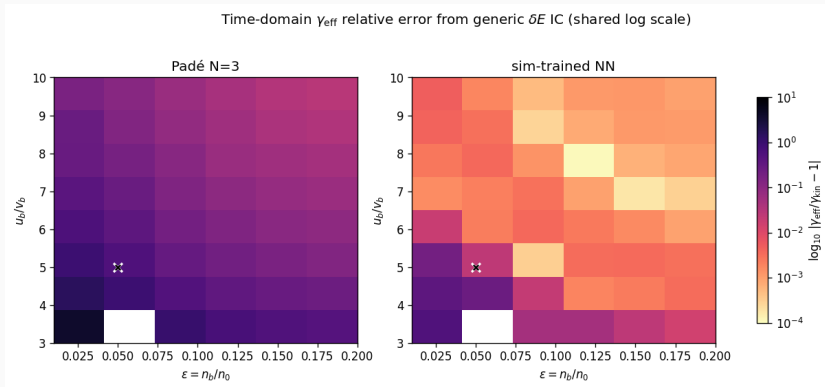
$\gamma(k)$ tracks kinetic through the unstable branch at 3/4 representative points; $\lesssim 1\%$ peak overshoot. Marginal corner (3, 0.05) shows the first crack: +12%.

Sim-trained NN: canonical time-domain looks fine



At $(5, 0.05, k=0.24)$: tracks kinetic under both Langmuir-projected and generic- δE ICs. Fitted $\gamma_{\text{eff}} = +0.1701$ vs $\gamma_{\text{kin}} = +0.1700$. Canonical point is well inside training distribution.

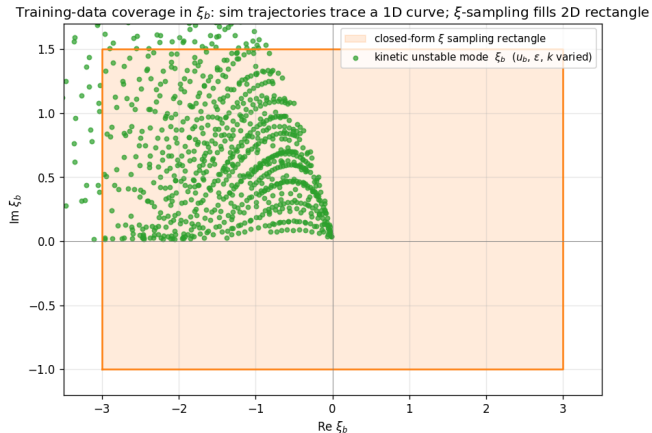
Sim-trained NN: 2D scan



$\log_{10} |\gamma_{\text{eff}}/\gamma_{\text{kin}} - 1|$, shared across all 2D sweeps. Sim-trained NN: one decade better than Padé in the well-trained interior, 10^{-2} – 10^{-1} most cells, $\sim 10^{-0.3}$ at the marginal corner.

Within $\sim 10\%$ on γ_{eff} over most of the grid — genuinely solid for 10^5 sim samples. Can we do better, cheaper?

Diagnosis: where is the data in ξ_b ?



ξ_b on the unstable kinetic mode as $(u_b/v_b, \epsilon, k)$ vary — a narrow 1D-ish curve in the upper-left quadrant. The closed-form ξ_b rectangle (orange) covers *all* ξ_b relevant for any fluid eigenmode (unstable + damped). Sim training never reaches the rest.

What the gradient is telling us

Generic δE IC excites *all five* fluid eigenmodes — Langmuir-like plus four others. Their ξ_b projections land all over the rectangle.

- Sim training sees only the unstable Langmuir mode's ξ_b .
- Other modes evaluated at moment ratios off the green curve $\Rightarrow$ NN extrapolates α .
- When extrapolation is benign (canonical, hot beam) the trajectory still settles to γ_{kin} ; when it isn't (marginal corner) the error grows.

The error gradient in the heatmap is the gradient of NN extrapolation quality. More simulations don't change it: trajectories live on the eigenmode manifold; their ξ_b -projection *is* the green region.

Fix: sample ξ_b directly

Moment ratios on a linear eigenmode are closed-form in ξ_b via the recurrence $U_{n+1} = \xi_b U_n - n M_{n-1} \hat{\Phi}$:

$$U_1/U_0 = \xi_b, \quad U_2/U_0 = \xi_b^2 - 1/Z'(\xi_b), \quad \alpha = \xi_b^3 - \xi_b/Z'(\xi_b)$$

One Faddeeva evaluation per training pair. No simulation required.

Sample uniformly: ξ_b in $\text{Re } \xi_b \in [-3, 3]$, $\text{Im } \xi_b \in [-1, 1.5]$. 16,384 samples.

Cost. $\sim 0.1\text{s}$ of `scipy.special.wofz` vs minutes of kinetic integration for $\sim 10^5$ sim samples — $\sim 100\times$ cheaper, no op-pt sweep, no ICs, no outlier filtering, no operating regime to choose.

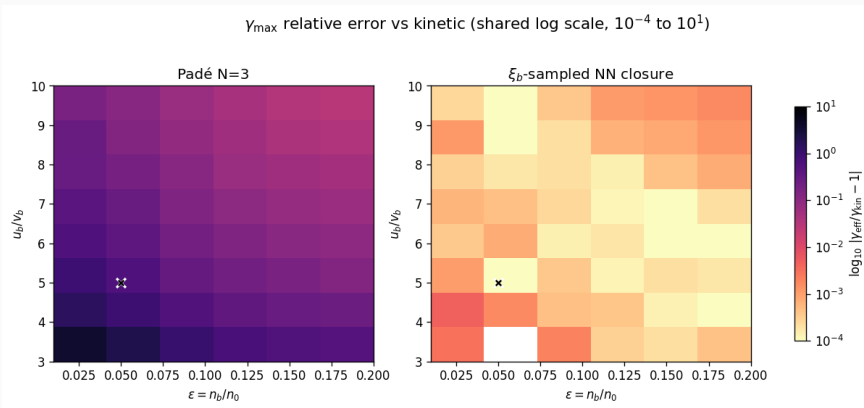
Same physics, same NN, same training schedule. Only the data changes.

Off-mode α accuracy: sim vs ξ_b -sampling

ξ_b	sim (1 op pt)	sim (broad, 24)	closed-form
$-1.17 + 0.71i$ (eigenmode)	+2.3%	+0.8%	+0.05%
$-0.83 + 0.05i$	+148%	+62%	+0.5%
$0.50 + 0.00i$	+843%	+247%	+0.6%
$-1.50 - 0.20i$	+50%	+27%	+0.2%
$1.20 + 0.10i$	+209%	+118%	+0.6%
$-2.00 + 0.00i$	+13%	+7%	+0.5%

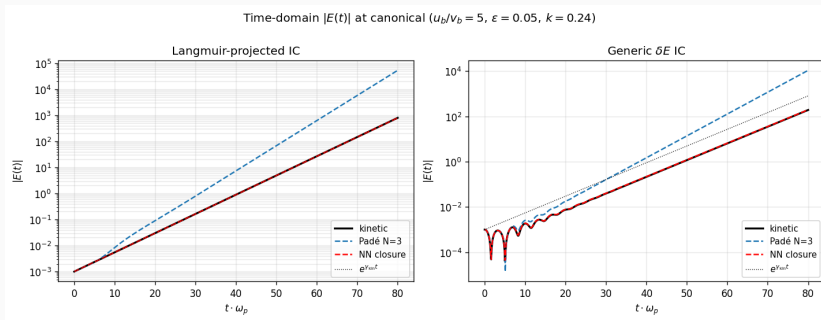
$|\alpha_{\text{NN}} - \alpha_{\text{exact}}|/|\alpha_{\text{exact}}|$ at test ξ_b . Sim training fits only the unstable curve; ξ_b -sampling is uniformly accurate.

$\gamma_{\max}$ sweep: ξ_b -sampled NN vs Padé



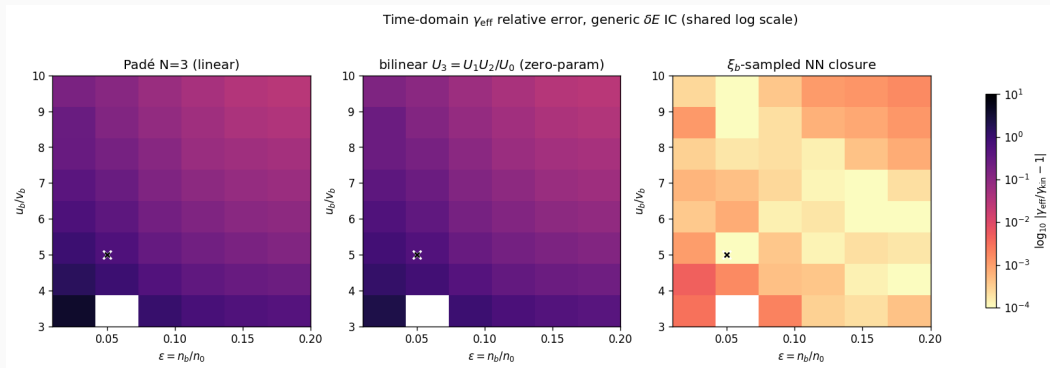
Same log scale. Padé $N=3$ sits at 10^{-1} – 10^0 ; ξ_b -sampled NN is $\leq 10^{-3}$ nearly everywhere. Two decades better than sim-trained, three to four decades better than Padé.

Time-domain stress test (ξ_b -sampled NN)



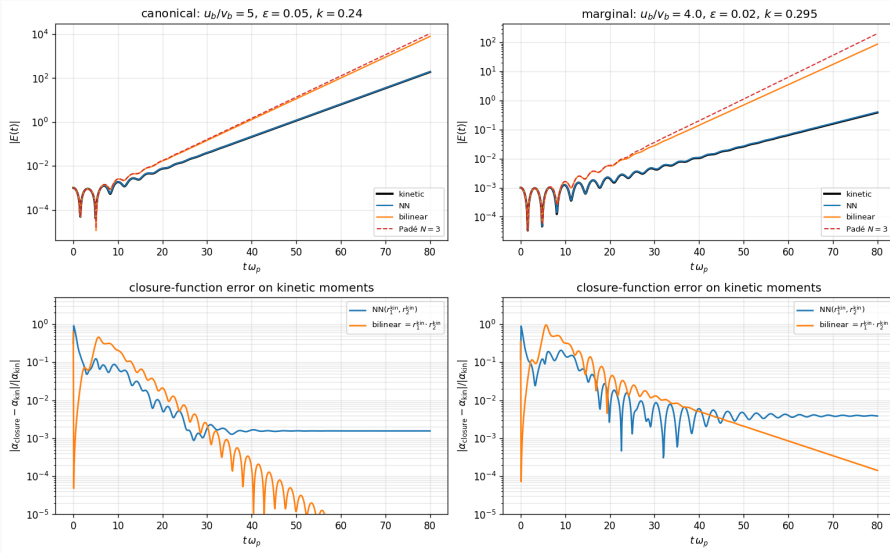
Generic δE IC, (5, 0.05, $k=0.24$). Fitted γ_{eff} (Langmuir / generic): kinetic +0.1700/ + 0.1700, Padé +0.2223/ + 0.2223, ξ_b -sampled NN +0.1700/ + 0.1700.

Time-domain 2D sweep: Padé, bilinear identity, NN



Padé and the bare bilinear identity are visually indistinguishable on the shared log scale — bilinear is a strict but small improvement ($\sim 10\%$ relative reduction at canonical, larger only at the marginal corner where both are already deep into failure). The ξ_b -sampled NN is two to three decades tighter across the entire grid. *Being structurally correct per-mode is not enough; the off-manifold behaviour is what makes the NN work.*

What is the NN doing? Off-manifold diagnostic



Top: $|E(t)|$. Bilinear (orange) tracks Padé, not kinetic. **Bottom:** closure-function error $|\alpha_{\text{closure}} - \alpha_{\text{kin}}|/|\alpha_{\text{kin}}|$ evaluated on

What did the NN learn?

Off-manifold check: evaluate $f_\theta(r_1, r_2)$ at (r_1, r_2) drawn *independently* from the on-manifold bounding box (not consistent with any single ξ_b).

sample location	$ f_\theta - r_1 r_2 / r_1 r_2 $ (mean)
on eigenmode manifold (training)	3×10^{-3}
off manifold (independent r_1, r_2)	~ 1

The NN did *not* learn complex multiplication globally. It learned a particular smooth interpolation in (r_1, r_2) -space that

- matches the bilinear product on the 2-real-d eigenmode submanifold (where it was trained), and
- happens, off-manifold, to suppress superposition cross-term error.

Per-mode, the structurally correct closure is bilinear (provable). *On a real trajectory*, the NN's smooth off-manifold extension is what makes the closure stable. Hardcoded multiplication is the wrong extension; the NN found a useful one.

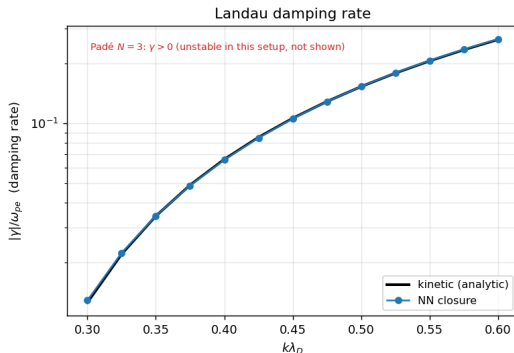
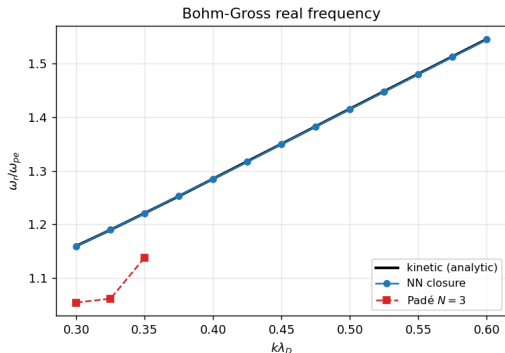
Generalization: same NN, Landau damping

The trained closure depends only on the Maxwellian linear response — no u_b , ε , k in the training data. Should work for any 3-moment Maxwellian-fluid setup. Test: Langmuir wave on a single warm Maxwellian, no beam, no cold bulk. Strip BoT to:

$$\partial_t E = -U_1, \quad \partial_t U_0 = -ikU_1, \quad \partial_t U_1 = -ikU_2 + E, \quad \partial_t U_2 = -ikU_3.$$

Same checkpoint. No retraining.

Landau damping on a single warm Maxwellian (same NN as BoT, no retraining)



Headline

Closure result. ξ_b -sampled NN gives 10^{-4} – 10^{-3} relative error on γ_{eff} across the $(u_b/v_b, \varepsilon)$ grid — two decades tighter than sim-trained, three to four decades tighter than Padé. Data is 16,384 Faddeeva evaluations, $\sim 100\times$ cheaper than sim sampling, problem-agnostic.

Generalization. Same checkpoint, no retraining, reproduces Landau damping on a single warm Maxwellian to $< 1\%$ across $k\lambda_D \in [0.3, 0.6]$. The closure object is a learned Maxwellian linear-response function, reusable for *any* 3-moment fluid integrator on a Maxwellian equilibrium (BoT, Langmuir damping, ion-acoustic, drift waves).

Interpretability. A one-line algebraic identity from the recurrence $+M_1=0$ pins the exact per-mode closure to a complex product, $\alpha = (U_1/U_0)(U_2/U_0)$. The NN reproduces this on the eigenmode manifold (where it was trained) but does *not* learn multiplication globally — off the manifold it produces a smooth interpolation that is distinctly non-multiplicative. Hardcoding multiplication directly produces a closure that fails almost as badly as Padé on generic ICs (cross-terms break the identity). The off-manifold smooth extension — not the on-manifold identity — is what makes the NN closure work.

Same NN. Strictly better data.

Mechanism: per-mode product on the manifold, useful smooth extension off it.